

Assignment 1

Part 1 - sets

Ans 1.1 $A = \{x | x \in \mathbb{R}, -1 \leq x \leq 2\} = [-1, 2]$

$$B = \{x | x \in \mathbb{R}, 1 \leq x \leq 4\} = [1, 4]$$

(a) $A \cup B = [-1, 4]$

(b) $A \cap B = [1, 2]$

(c) $A \setminus B = [-1, 1)$

(d) $\bar{A} \cap B$

$$\bar{A} = (-\infty, -1) \cup (2, \infty)$$

$$\bar{A} \cap B = (2, 4]$$

~~Ans 2. We have two biased coins,~~

Part 2 - Probability

Ans 2.1 two coins a and b are tossed simultaneously,

$$P(H_a) = 0.6 \quad P(H_b) = 0.4$$

(a) Sample space of the experiment

$$\Omega = \{H_a H_b, H_a T_b, T_a H_b, T_a T_b\}$$

(b) Event E_1 : heads appear on either coin a or coin b or both.

$$E_1 = \{H_a H_b, H_a T_b, T_a H_b\}$$

Event E_2 : heads appear only on coin a.

$$E_2 = \{H_a T_b\}$$

(i) $P(E_1) = P(H_a H_b) + P(H_a T_b) + P(T_a H_b)$

$$= P(H_a) \cdot P(H_b) + P(H_a) \cdot P(T_b) + P(T_a) \cdot P(H_b)$$

since, outcomes of both coins are independent

$$= 0.6 \times 0.4 + 0.6 \times 0.6 + 0.4 \times 0.4$$

$$= 0.24 + 0.36 + 0.16$$

$$= 0.76$$

(ii) $P(E_2) = P(H_a T_b)$

$$= P(H_a) \cdot P(T_b)$$

$$= 0.6 \times 0.6$$

$$= 0.36$$

(c) Probability of occurrence of each elementary event in Ω

$$P(H_1 H_2) = P(H_1) \times P(H_2) = 0.6 \times 0.4 = 0.24$$

$$P(H_1 T_2) = P(H_1) \times P(T_2) = 0.6 \times 0.6 = 0.36$$

$$P(T_1 H_2) = P(T_1) \times P(H_2) = 0.4 \times 0.4 = 0.16$$

$$P(T_1 T_2) = P(T_1) \times P(T_2) = \cancel{0.4 \times 0.4} 0.4 \times 0.6 = 0.24$$

Ans 2.2 Margin (cm): $\{5.76, 4.56, 2.33, 6.32, 5.4\}$

Sample Space $\Omega = \{(\omega_1, \omega_2, \omega_3, \omega_4, \omega_5) \mid \omega_i \in [0, \infty)\}$

For the observed experiment, the given outcome is:

$$\omega_0 = (5.76, 4.56, 2.33, 6.32, 5.4)$$

one of the elements of Ω .

The problem defines -

→ Success, if min distance ≤ 5 cm

→ Failure, if min distance > 5 cm

(a) Random variable 'X' counts the number of success in S trials.

$$X: \Omega \rightarrow \Omega_X$$

$$\Omega_X = \{0, 1, 2, 3, 4, 5\}$$

Now, we have to define X as a function of ω

$$\omega = (\omega_1, \omega_2, \omega_3, \omega_4, \omega_5) \in \Omega$$

$$\phi(\omega_i) = \begin{cases} 1, & \text{if } \omega_i \leq 5 \text{ (success)} \\ 0, & \text{if } \omega_i > 5 \text{ (failure)} \end{cases}$$

$$X(\omega) = \sum_{i=1}^S \phi(\omega_i)$$

(b) Probability distribution of X .

→ Each shot is independent

→ Probability of success in one shot = p

$$P(X = k) = {}^S C_k p^k (1-p)^{S-k} ; k \in \Omega_X$$

where, k is the number of successes in S trials.

Part 3: Discrete random variables

Am 3.1 Random Variable X = number of words correctly recognized out of 50.

Number of trials = 50

Probability of success $(p) = 0.9$

$$P(X=k) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

We are given, $k=45$, $n=50$, $p=0.9$

$$P(X=45) = \frac{50!}{45! 5!} (0.9)^{45} (0.1)^5$$

$$P(X=45) = 0.1849246$$

Am 3.2 $f(k, \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$; $\lambda=10$

$$P(k=0) = \frac{10^0 e^{-10}}{0!} = e^{-10} = 0.0000454$$

The rate of COVID infections

4.1 What is the probability that 100 infections occur in a day?

```
import math
import numpy as np
import matplotlib.pyplot as plt
```

```
lam=10
#Probability Mass Function
def pmf(k):
    return (lam**k * math.exp(-lam)) / math.factorial(k)
```

```
# 4.1 Probability that 100 infections occur in a day
p_100 = pmf(100)
print("P(X = 100) =", p_100)
```

```
... P(X = 100) = 4.864649182067611e-63
```

What is the probability that the number of infections would be somewhere between 5 and 15?

```
# 4.2 Probability that infections are between 5 and 15 (inclusive)
p_5_15 = sum(pmf(k) for k in range(5, 16))
print("P(5 ≤ X ≤ 15) =", p_5_15)
```

```
P(5 ≤ X ≤ 15) = 0.9220069086190603
```

4.3 What is the sample space of covid infection rate (observation) i.e., what are possible values that can be reported as "number of infections in a day"?

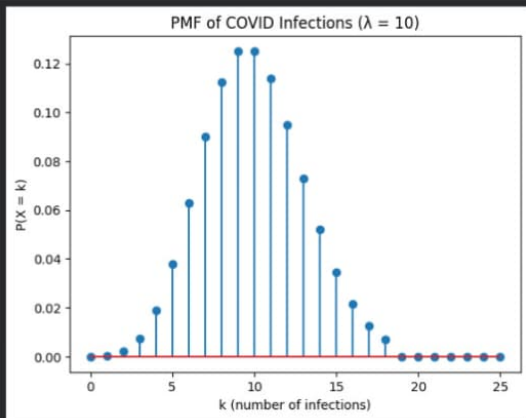
```
# 4.3 Sample space  
print("Sample space  $\Omega = \{0, 1, 2, 3, \dots\}$ ")
```

```
Sample space  $\Omega = \{0, 1, 2, 3, \dots\}$ 
```

4.4 Draw the graph of the probability mass function $f(k)$.

```
# 4.4 Plot the probability mass function
k_values = np.arange(0, 26)
pmf_values = [max(pmf(k), 0) for k in k_values]

plt.figure()
plt.stem(k_values, pmf_values)
plt.xlabel("k (number of infections)")
plt.ylabel("P(X = k)")
plt.title("PMF of COVID Infections ( $\lambda = 10$ )")
plt.show()
```



4.5 Suppose the actual infection count reported in the last ten days is 18, 20, 23, 25, 22, 30, 11, 10, 23, 12. Do you think the researcher's claim can be correct?

```
# 4.5 Observed infection counts
observed_data = np.array([18, 20, 23, 25, 22, 30, 11, 10, 23, 12])
print("Observed mean =", observed_data.mean())
```

```
Observed mean = 19.4
```

The mean of given data is 19.4 and not 10. The researcher's claim is not likely to be correct.

4.6 Generative process

```
# 4.6 Generative process (Poisson random generator)
def generative_process(n=1):
    return np.random.poisson(lam, size=n)
```

```
# 4.6.1 Generate predictions for next 5 days
predictions_1 = generative_process(5)
print("First 5-day prediction:", predictions_1)
```

```
# 4.6.2 Generate predictions again for next 5 days
predictions_2 = generative_process(5)
print("Next 5-day prediction:", predictions_2)
```

```
First 5-day prediction: [ 7  7  9 12 18]
Next 5-day prediction: [12 10  2 14 10]
```

The new 5 predictions are not the same as previous 5 predictions.